Sensitivity Analysis with Respect to the Choice of k.

This folder contains eight figures illustrating the sensitivity of the proposed IF-CPD to the choice of k.

Specifically, the figures are organized into two categories:

Subspace settings (6 figures):

For each of the two subspace scenarios (SPD subspace and probability distributional subspace), we provide three figures.

Each figure corresponds to a pair of sample sizes, namely n = (400,600), (800,1000), (1200,1600).

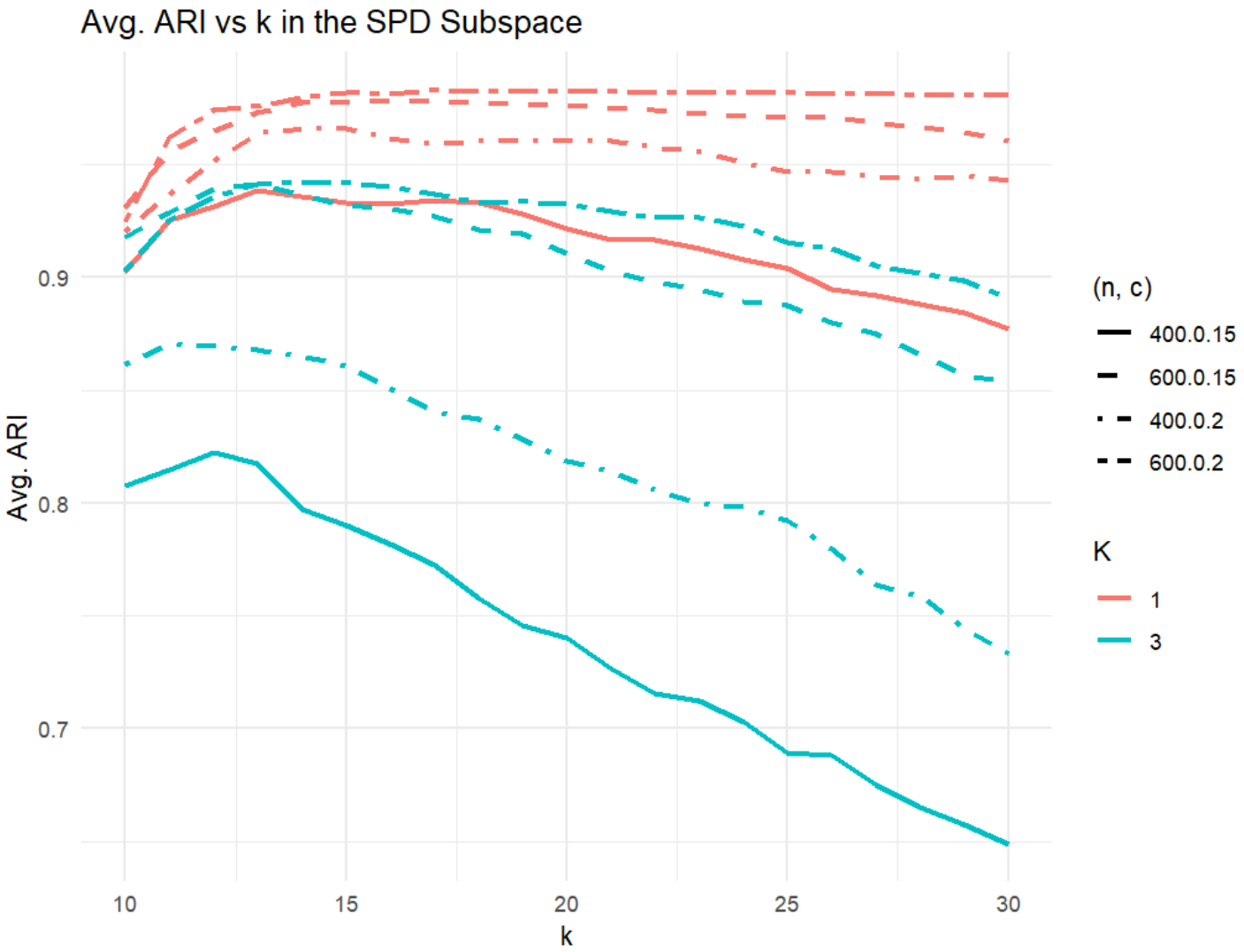
Each figure shows the curves of ARI (Adjusted Rand Index) as a function of k.

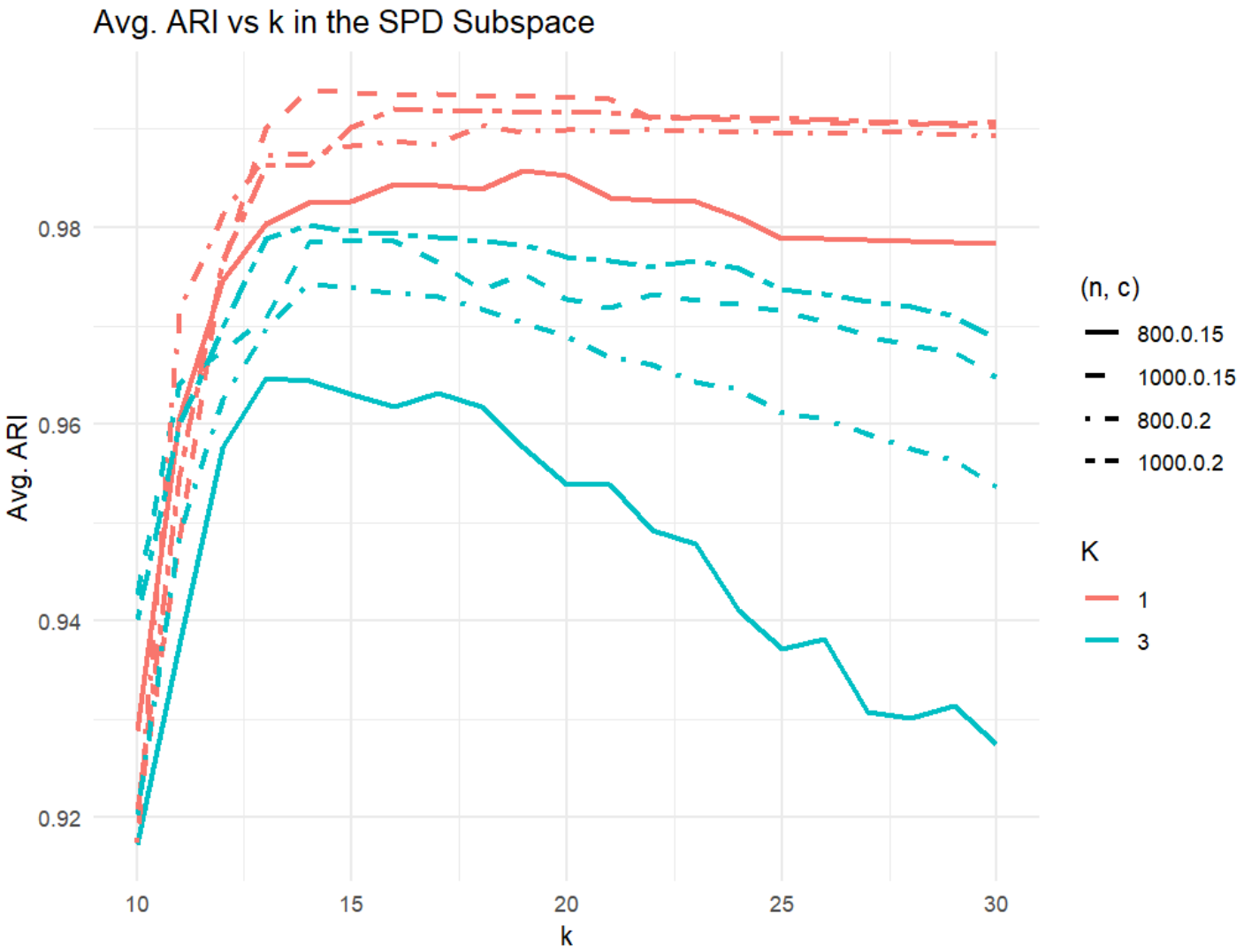
Full space settings (2 figures):

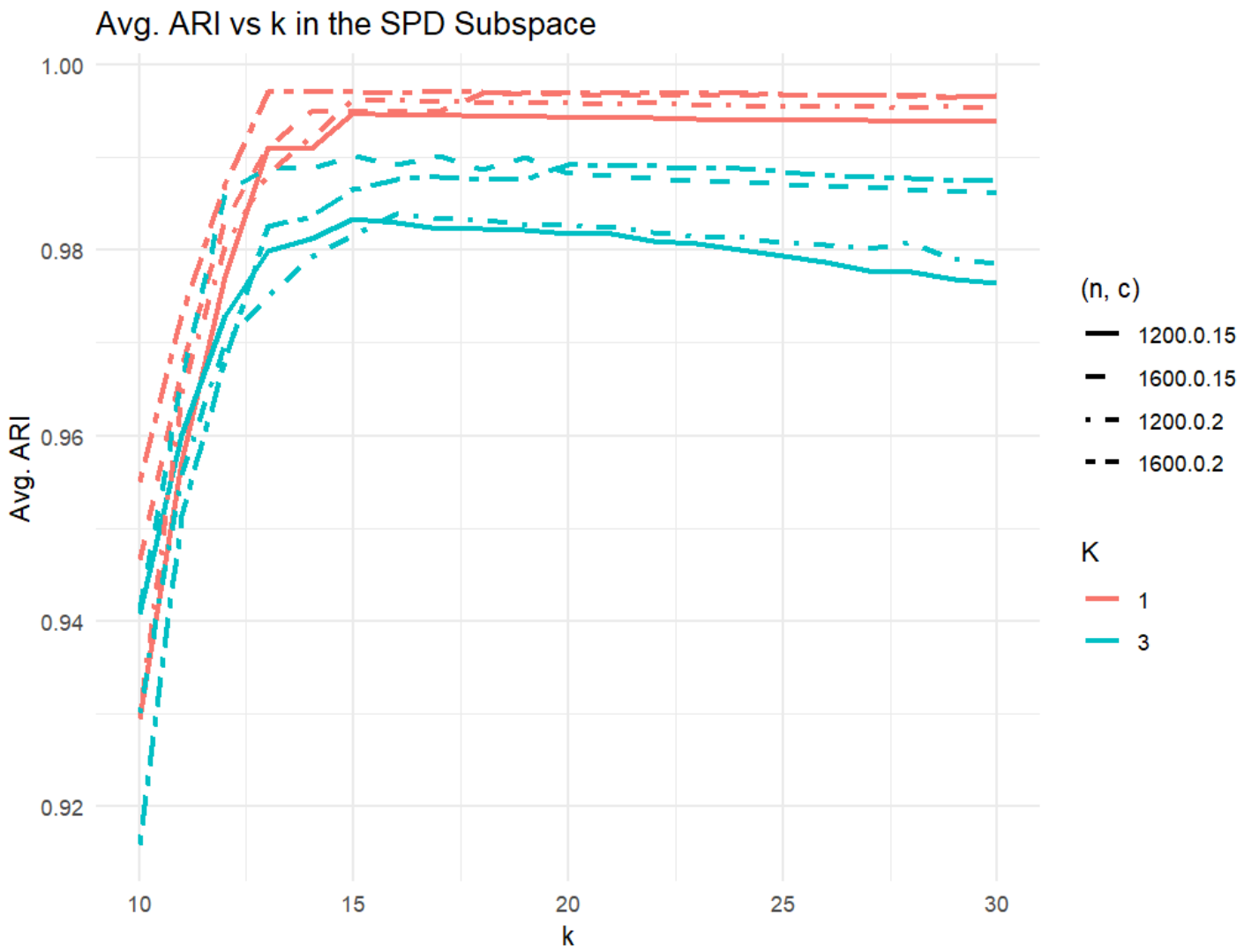
For each of the two full-space scenarios (SPD space and probability distributional space), we provide one figure corresponding to sample sizes n = (400,600).

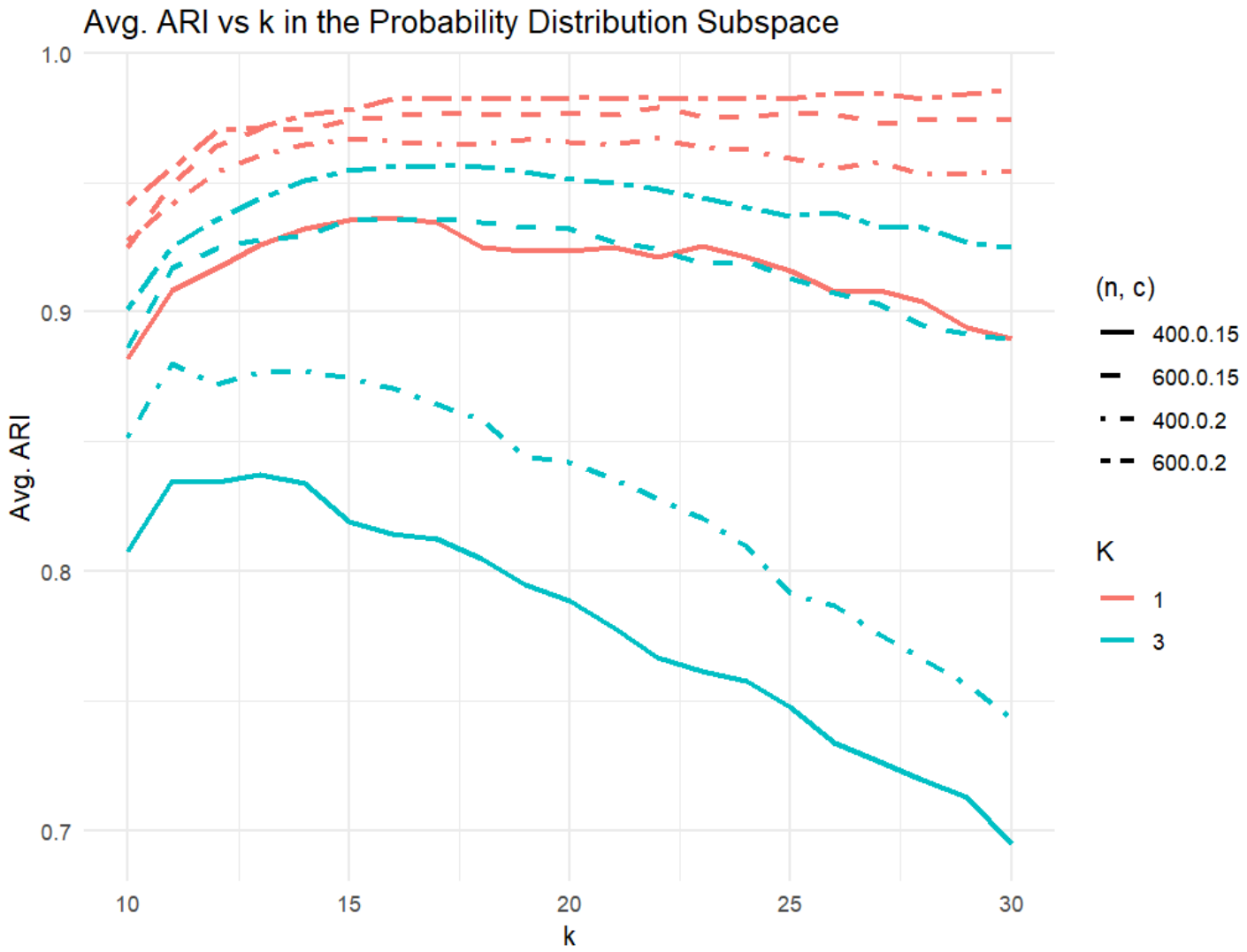
Each figure displays the average ARI as a function of k, computed over 500 Monte Carlo simulations.

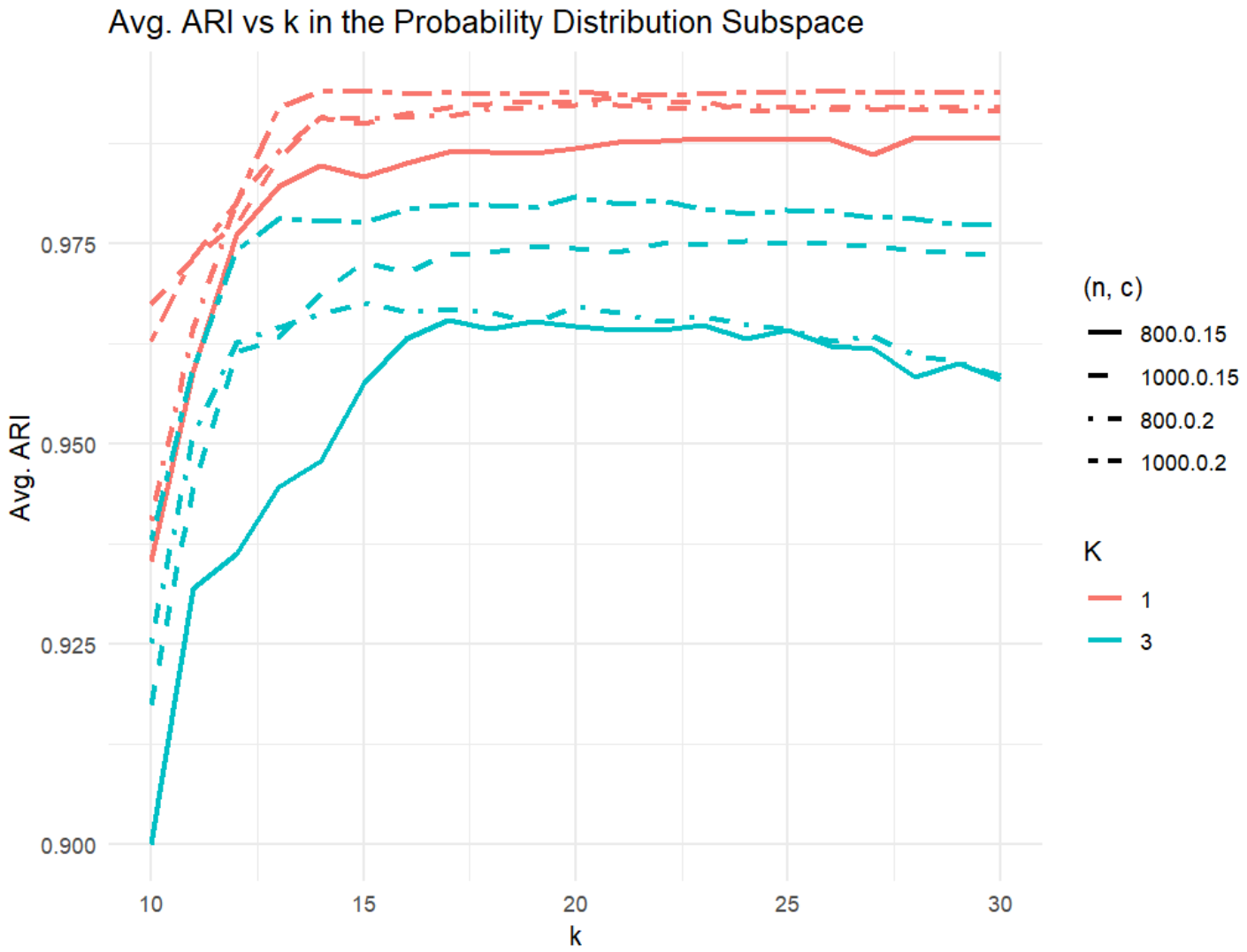
In each of the eight figures, eight curves are displayed, corresponding to two sample sizes (n), two numbers of change points (K), and two spacings between change points (c).

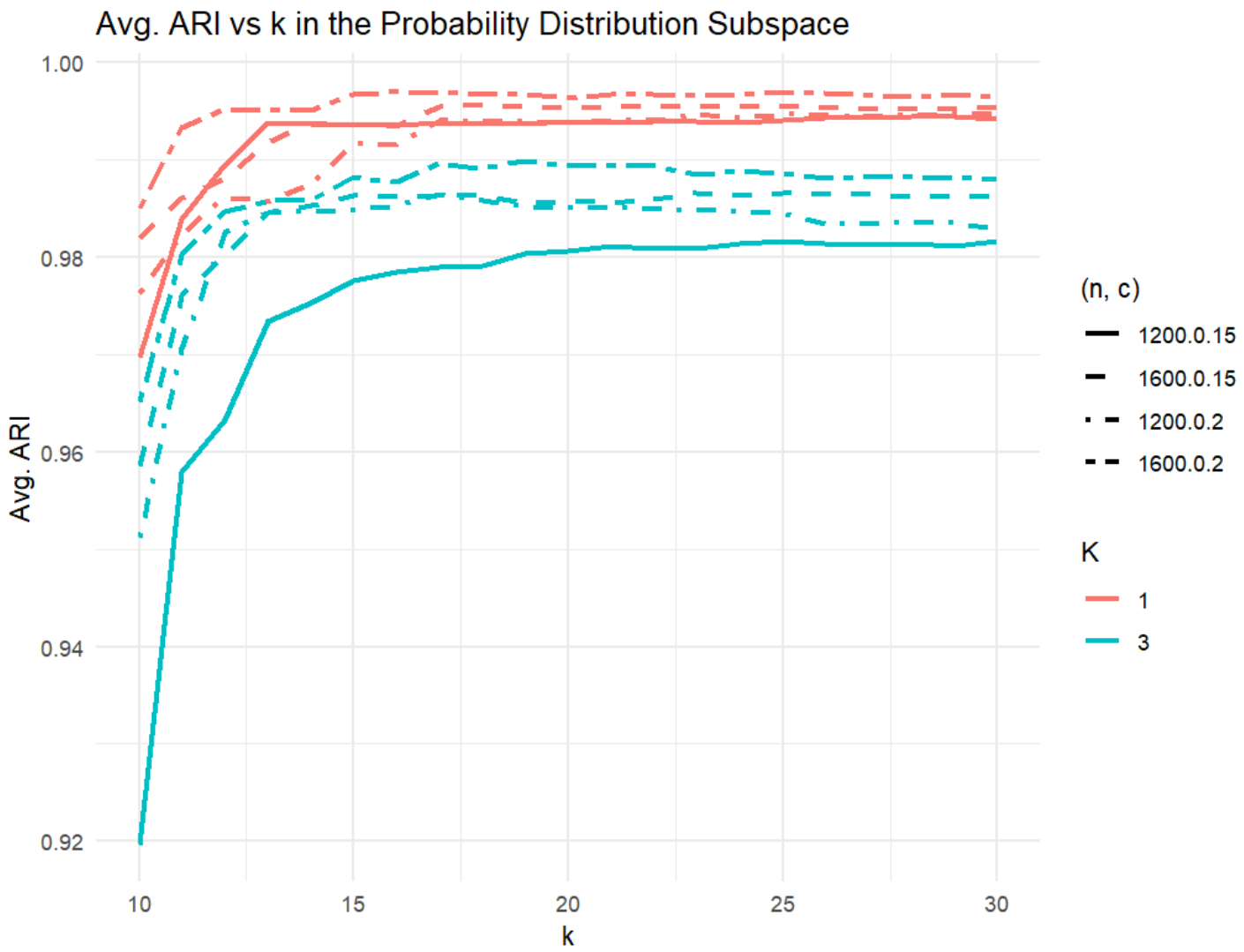

Avg. ARI vs k in the SPD Subspace
Avg. ARI
0.9
0.8
0.7
10
15
20
25
30
k
(n, c)
400.0.15
600.0.15
400.0.2
600.0.2
K
1
3

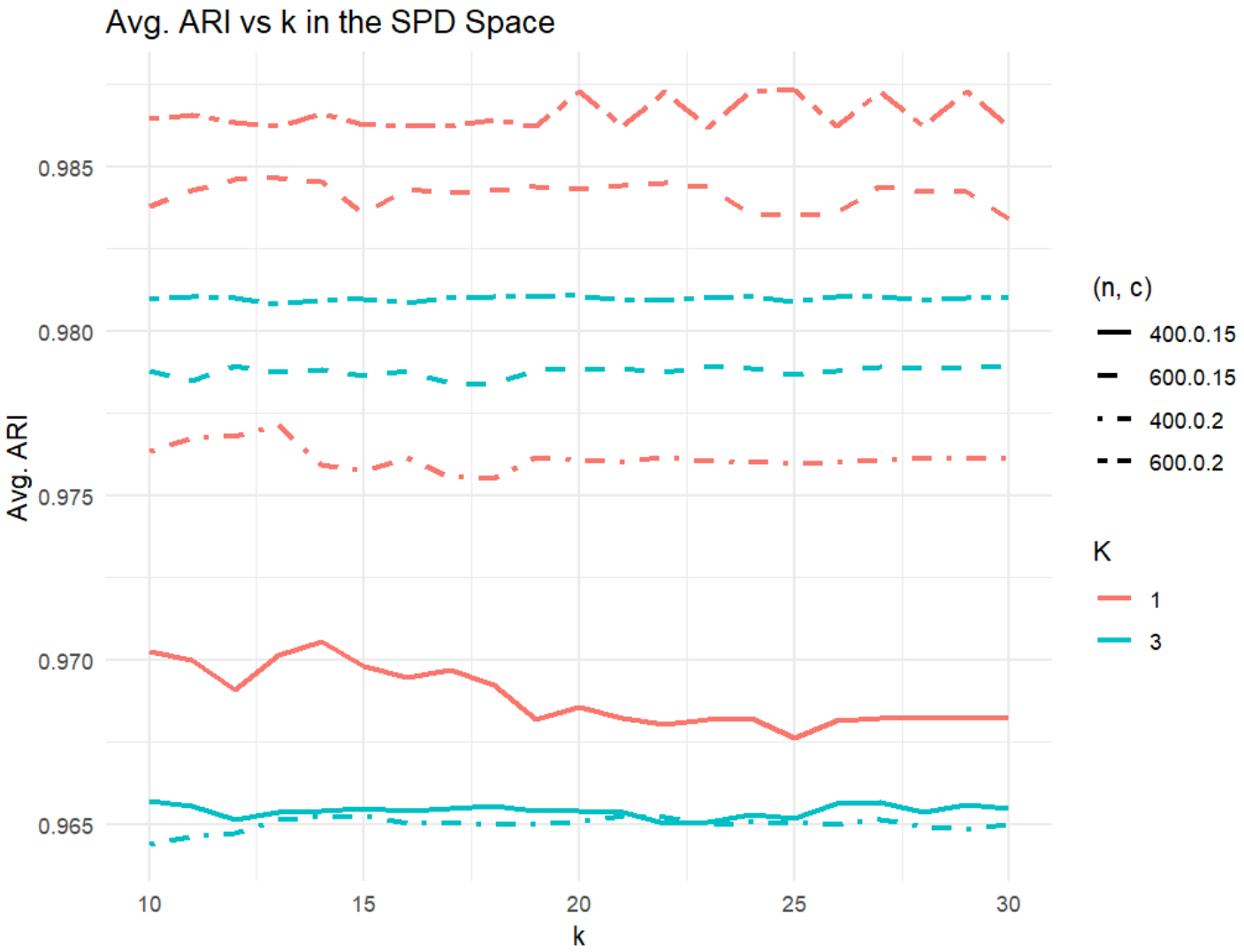

Avg. ARI vs k in the SPD Subspace
Avg. ARI
0.98
0.96
0.94
0.92
10
15
20
25
30
k
(n, c)
800.0.15
1000.0.15
800.0.2
1000.0.2
K
1
3

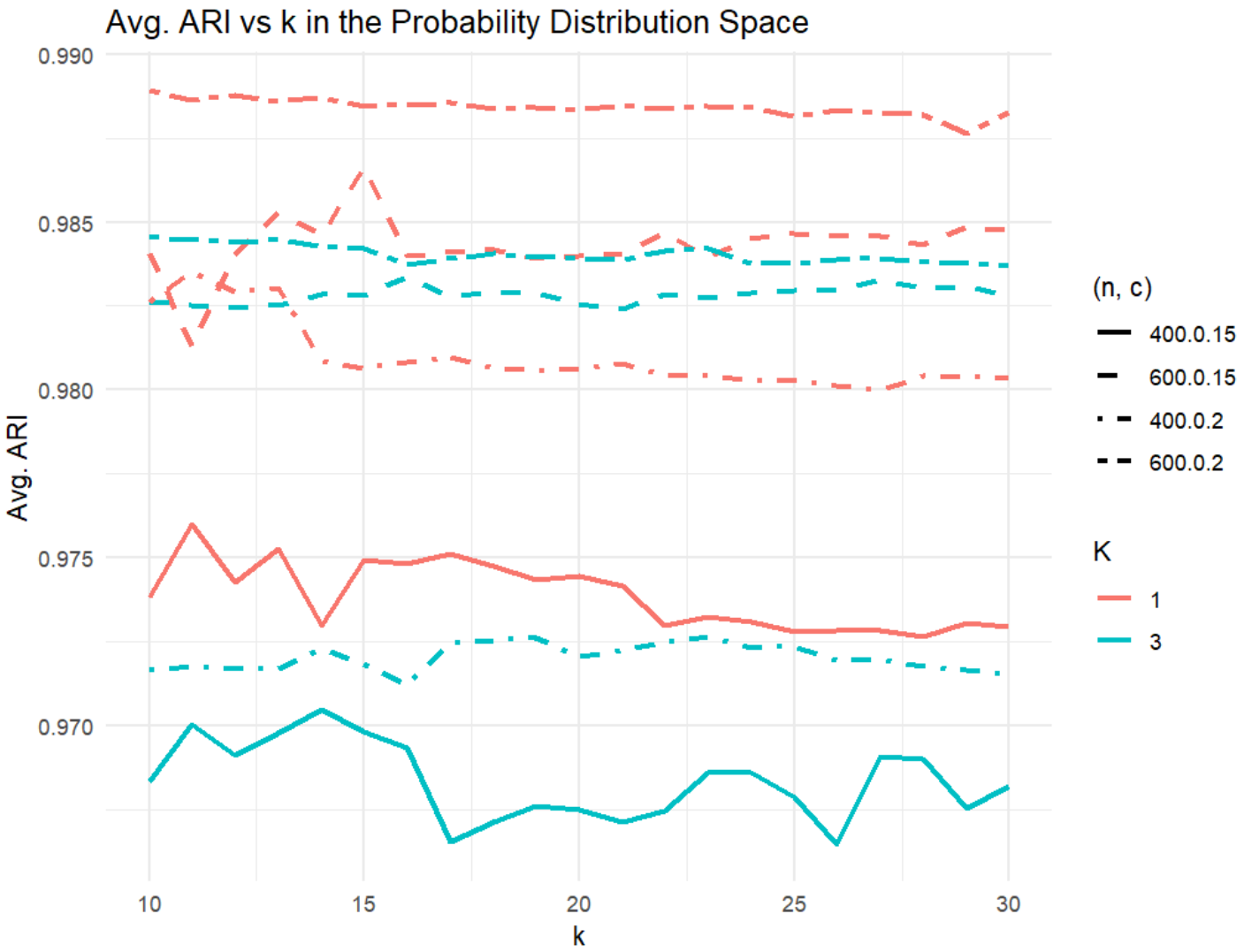

Avg. ARI vs k in the SPD Subspace
1.00
0.98
0.96
0.94
0.92
Avg. ARI
10
15
20
25
30
k
(n, c)
1200.0.15
1600.0.15
1200.0.2
1600.0.2
K
1
3


Avg. ARI vs k in the Probability Distribution Subspace
1.0
0.9
0.8
0.7
Avg. ARI
10
15
20
25
30
k
(n, c)
400.0.15
600.0.15
400.0.2
600.0.2
K
1
3


Avg. ARI vs k in the Probability Distribution Subspace
Avg. ARI
0.975
0.950
0.925
0.900
10
15
20
25
30
k
(n, c)
800.0.15
1000.0.15
800.0.2
1000.0.2
K
1
3


Avg. ARI vs k in the Probability Distribution Subspace
1.00
0.98
0.96
0.94
0.92
Avg. ARI
10
15
20
25
30
k
(n, c)
1200.0.15
1600.0.15
1200.0.2
1600.0.2
K
1
3


Avg. ARI vs k in the SPD Space
Avg. ARI
0.985
0.980
0.975
0.970
0.965
10
15
20
25
30
k
(n, c)
400.0.15
600.0.15
400.0.2
600.0.2
K
1
3


Avg. ARI vs k in the Probability Distribution Space
0.990
0.985
0.980
0.975
0.970
Avg. ARI
10
15
20
25
30
k
(n, c)
400.0.15
600.0.15
400.0.2
600.0.2
K
1
3